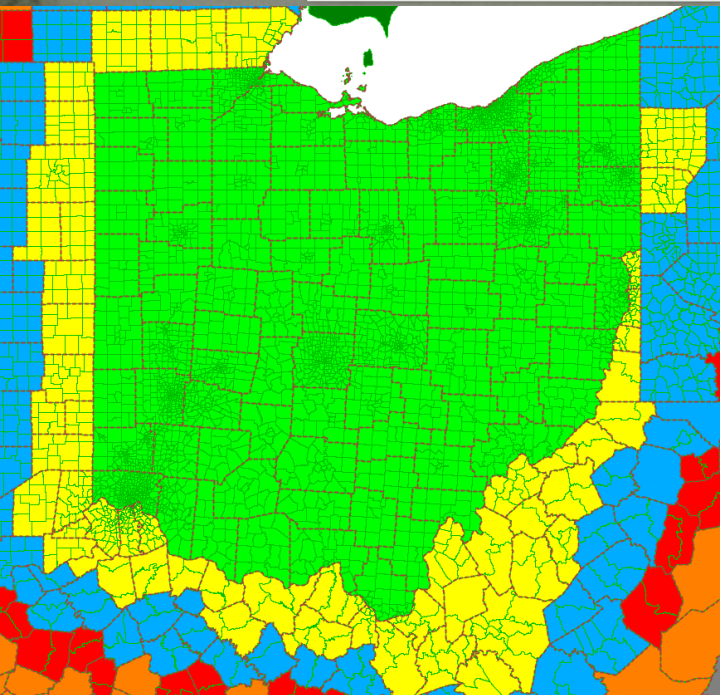


Various uses for INRIX/Streetlight data: Ohio plus border area

Sam Granato
For OTDMUG, 9-8-17



Project & Data Source types:

- (Project type #6 is AADT estimation)

O-D Analysis: Analysis of trips that start or initially pass through the Origin Zones and end or later pass through the Destination Zones. The Project must have at least one Origin Zone Set and at least one Destination Zone Set, and may also have Filter Zone Sets.

O-D Analysis with Middle Filter: Analysis of trips that start or initially pass through the Origin Zones, then pass through the Middle Filter Zones, and then end or pass through the Destination Zones. The Project must have at least one Origin Zone Set, at least one Destination Zone Set, and at least one Middle Filter Zone Set where all the Zones are marked as pass through. The Project may also have Filter Zone Sets.

O-D to Pre-set Geography: Analysis of trips between each Zone in one or more Zone Sets, and a standard Geography such as Zip Codes, Census Block Groups, or Traffic Analysis Zones (TAZ). The analysis includes using the Zone(s) as origin and Geography as destination, as well as Geography as origin, and the Zone(s) as destination. The project must have at least one Zone Set, and a selected Geography.

Zone Activity Analysis: Analysis of trips that start/end in or pass through each Zone in one or more Zone Sets. Project must have at least one Zone Set.

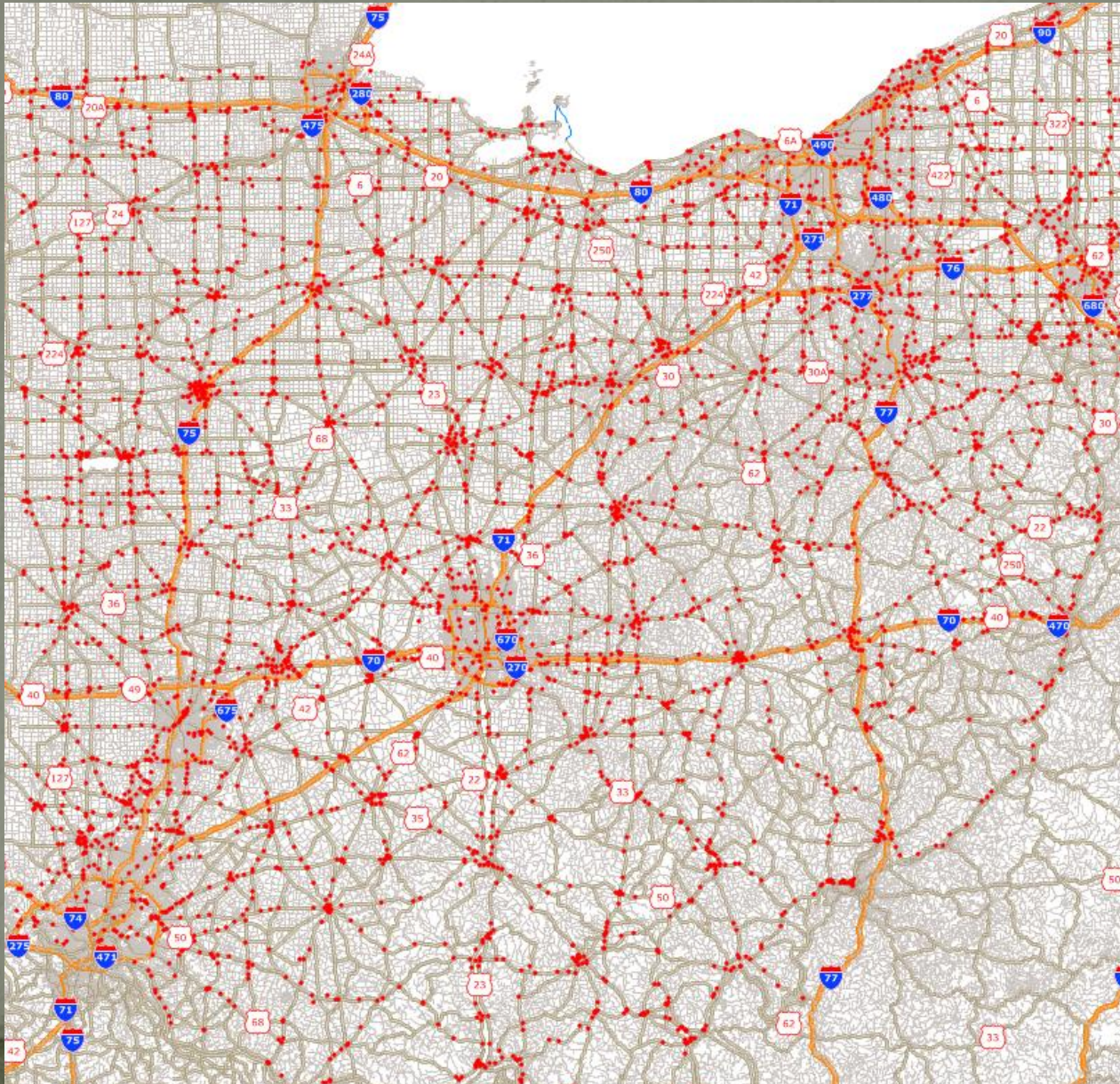
Visitor Home Work Analysis: Analysis of distribution of home and work locations of the visitors to each Zone, by segmentation such as census block groups, zip codes, 1 km square grid, metro areas, and U.S. states. The Project must have at least one Zone Set.

Navigation-GPS: Data collected from GPS devices (mobile phones, connected cars, truck tracking systems), typically when navigating on a road network. This data generally represents "vehicle trips" between start and end points.

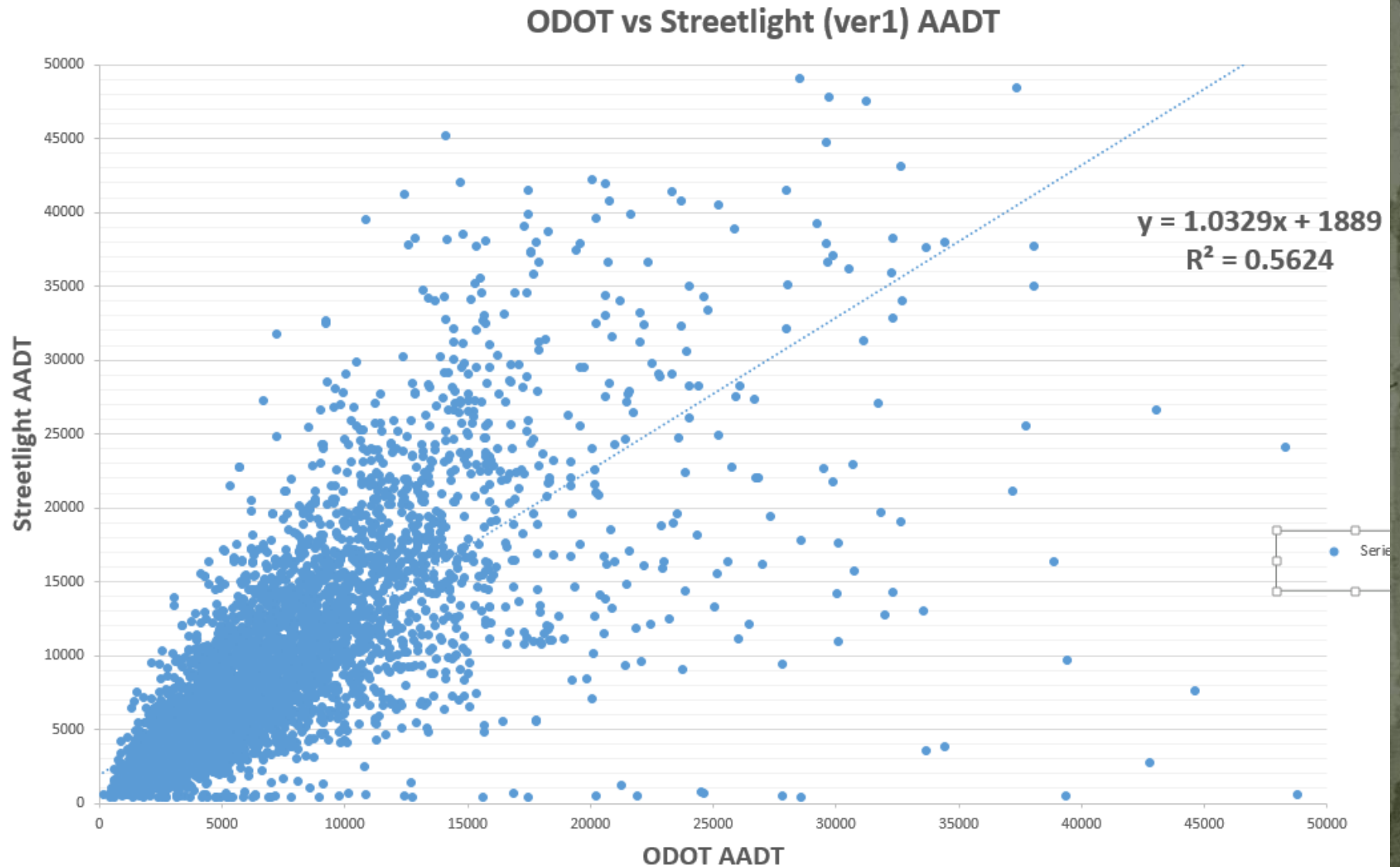
Location-Based Services: Data collected via smart-phone applications that use location data to provide their services (e.g. weather notifications when a storm is approaching the users' location). This data typically represents "user activity" at given

: Blend of data
mobile
k tracking
one
i data to
eather
:ation).

Streetlight AADT estimation: using mix of GPS vehicle sensors and smart phones (when using service apps)



AADT Prediction: So far found decent correlation, but 30% high overall - and not improved by the amount of volume.

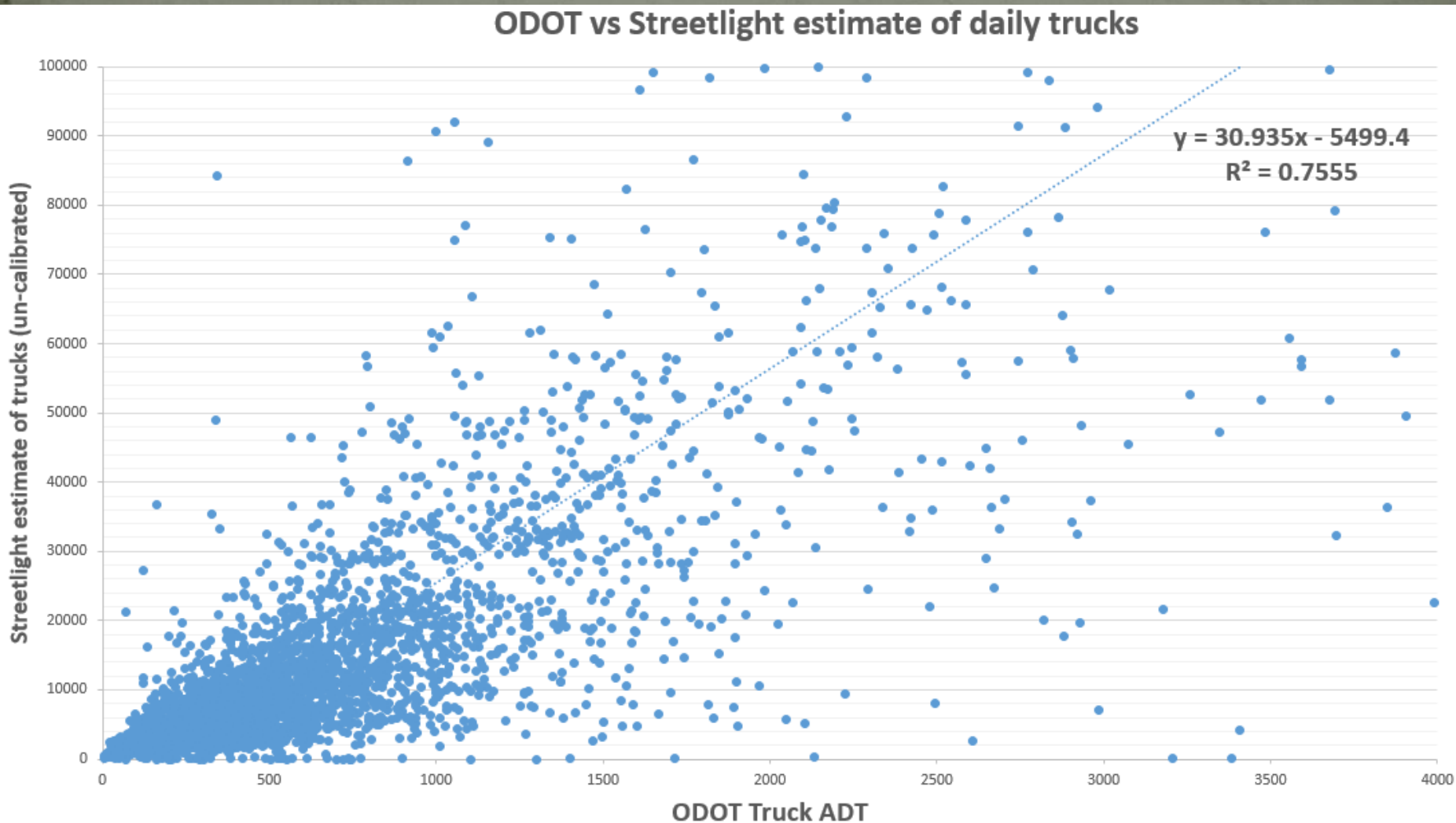


AADT estimates on rural collector/local roads (focus of ODOT effort to estimate volume/VMT on uncounted routes)

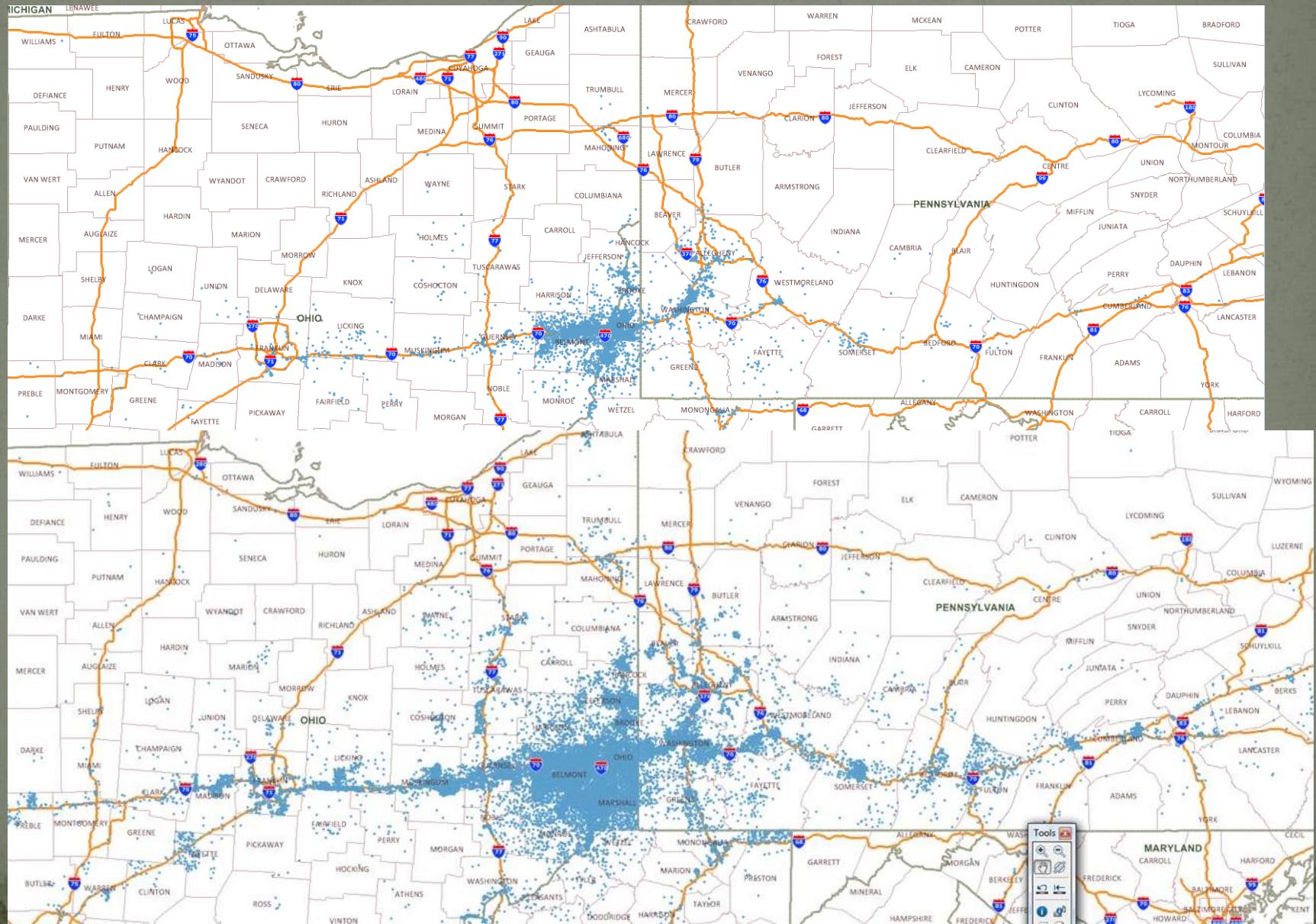
ODOT vs Streetlight (ver 1) AADT: Functional class 7-9 only



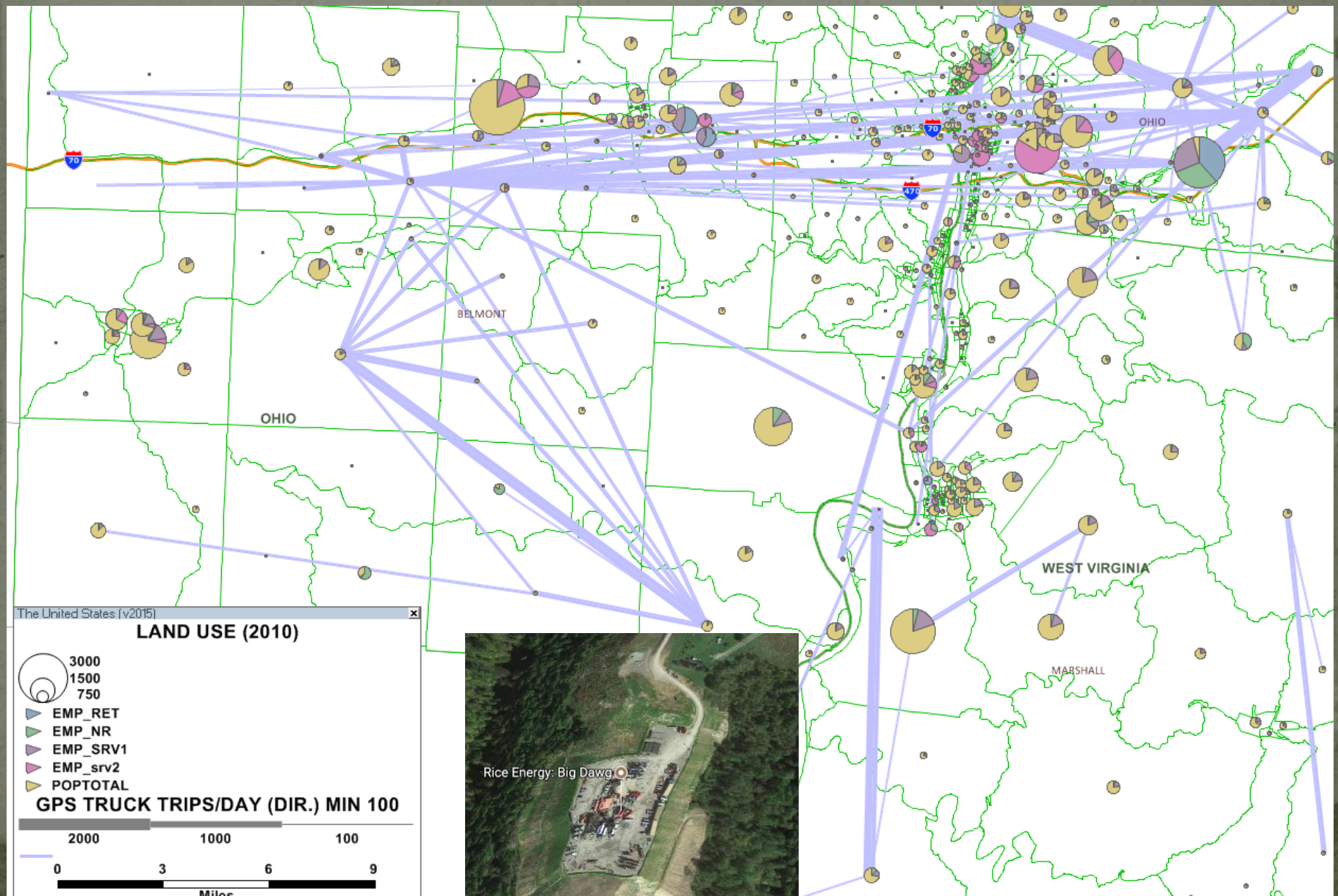
Used indirect means of estimating truck ADT and compared to ODOT count stations with truck ADT: (“Zone Activity analysis” with GPS data only)



Origins of traffic destined for the 3-county Wheeling area (using GPS data with “O/D to pre-set geography”) car (top) vs truck:



Estimated major zone-to-zone movements of “heavy trucks” per day. (GPS data with “O/D analysis”)

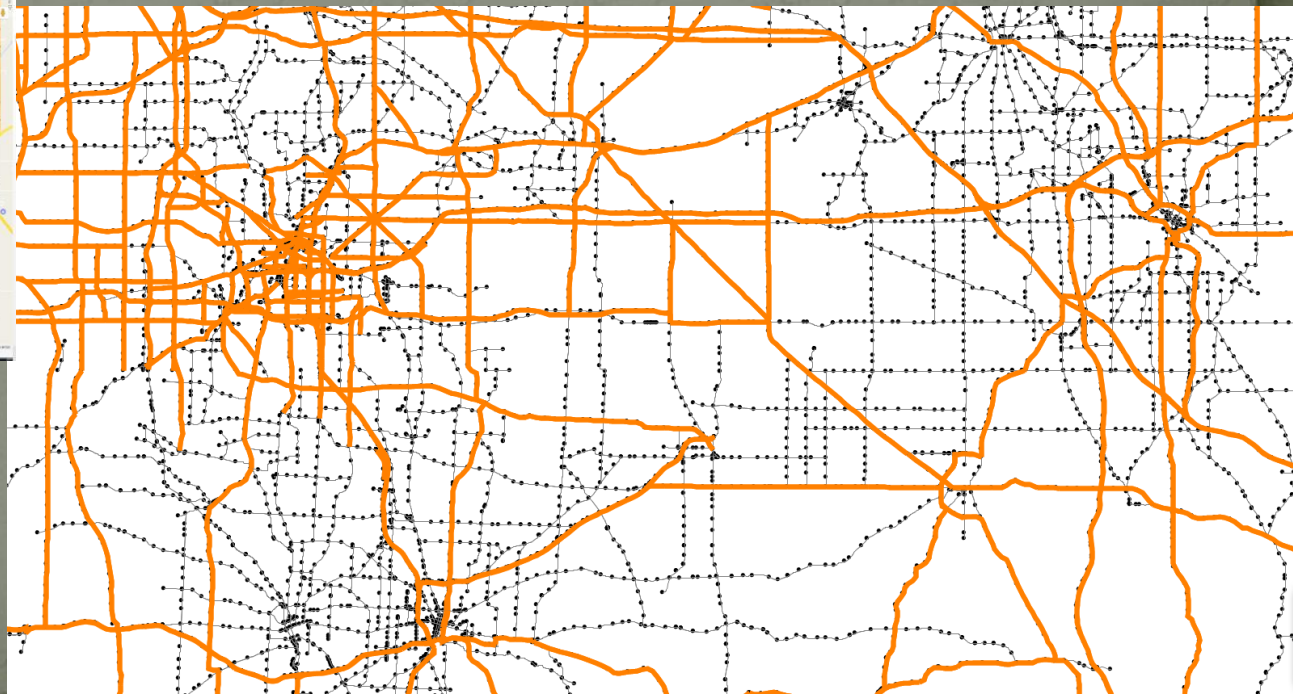
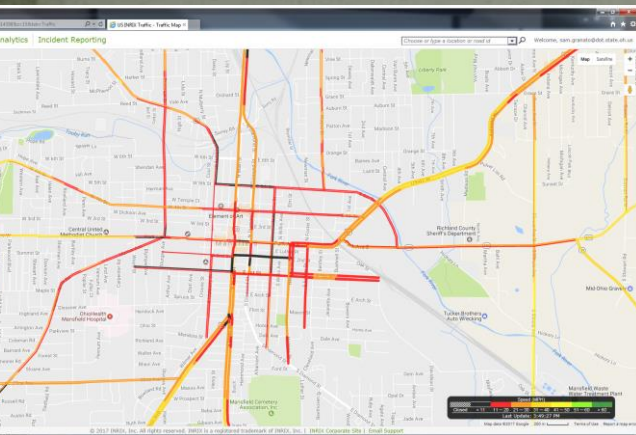


Inrix Trip waypoints (snapped to XD road net):



Beyond TMC - the INRIX XD network:

- Adds route coverage, esp. in areas below 500,000 pop (53,000 directional miles statewide @ 0.42 miles average length vs 35,000 miles @ 1.35 miles average length for the TMC network)
- I just reviewed new (re-segmented) historic average times on the XD network for 1 MPO vs the modeled travel times



Questions?

- sam.granato@dot.ohio.gov

